Ex no: 10A Design and configuration of a simple Internetwork using a Router

Aim:

To implement classless IP subtracting using cisco packet Tracer and configure the network devices with appropriate IP addresses and subnet masks to establish communication between different subnets

PROCEDURE

1. In cisco packet tracer create a new network topology

2. Add the required network devices

-> 2 routers (R0 and R_1)

-> 2 switches (S_0 and S_1)

-> To PCs (PC_5 - PC_4)

to switch S_1 using suitable ethernet cables

4. Connect S_0 to R_0 and S_1 to R_1

5. Connect R_0 and R_1 using their available fast Ethernet interfaces.

6. Divide the network into /27 subnets. The subnet mask used is: 255.255.255.224

7. Configure R_0 using the CLI. Assign 192.168.1.1 to the interface connected to S_0 and 192.168.1.65 to the interface connected to R_2

8. Enable the router interfaces using the no no shutdown command.

9. Configure R_1 using CLI. Assign 192.168.1.33 is the interface connected to S_1 and 192.168.1.66 to the interface connected to



Router0

Physical | Config | CLI | Attributes

```
Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

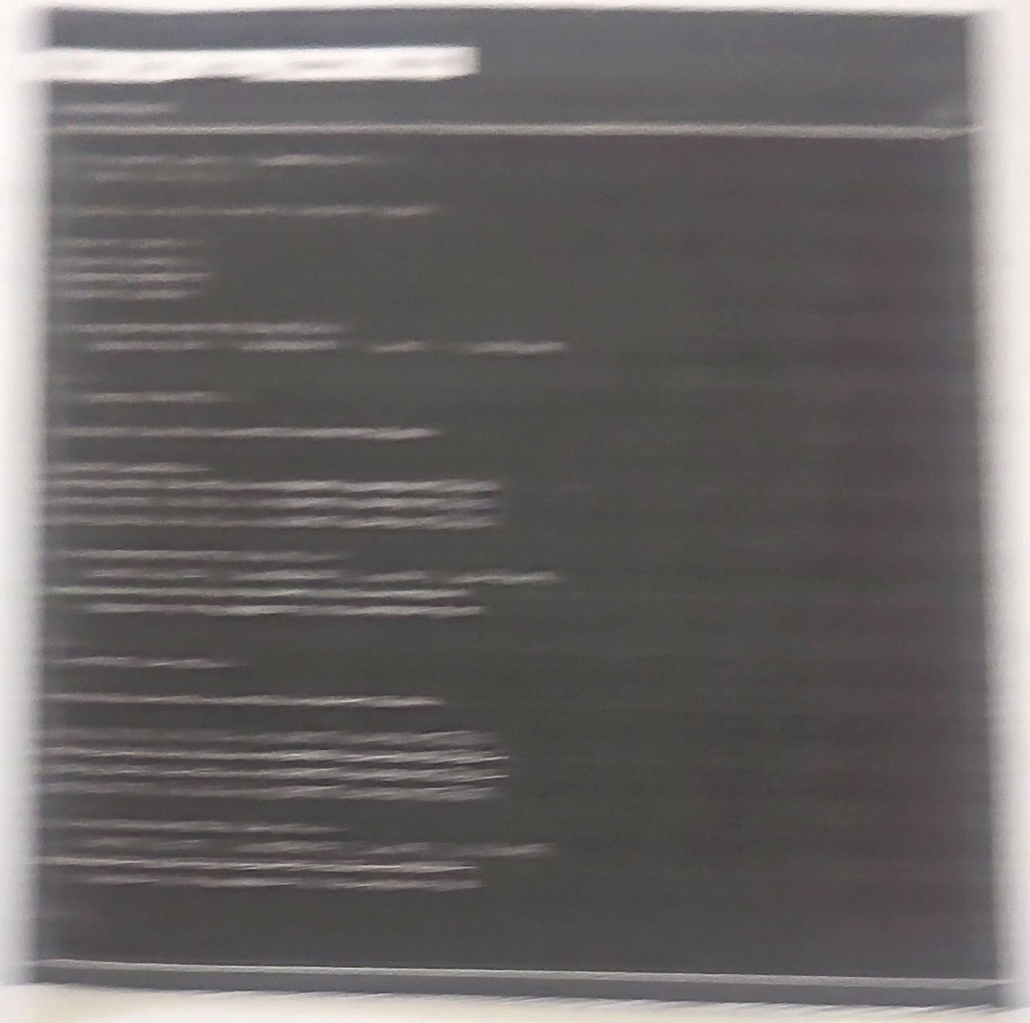
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    down
FastEthernet0/1        192.168.20.1    YES manual up                    down
Vlan1                  unassigned      YES unset  administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    up
FastEthernet0/1        192.168.20.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Copy | Paste





10. Enable the R1 interfaces using the no shutdown command in cmd.

11. configure the IP addresses of PC0 : PC4 in the 1st subnet

(a) PC0 - 192.168.1.11

(b) PC1 - 192.168.1.12

(c) PC2 - 192.168.1.13

(d) PC3 - 192.168.1.14

(e) PC4 - 192.168.1.15

use subnet mask 255.255.255.224 and default gateway 192.168.1.1

12. configure the IP addresses of PC5 - PC4 in the 2nd subnet.

(a) PC5 - 192.168.1.41

(b) PC6 - 192.168.1.42

(c) PC7 - 192.168.1.43

(d) PC8 - 192.168.1.44

(e) PC4 - 192.168.1.45

use subnet mask 255.255.255.224 and default gateway 192.168.1.33

13. Configure static routes on R0 and R1 so that packets can travel b/w the 2 lan subnets.

14. Verify the router interfaces using the command show ip interface brief

15. Test connectivity b/w PCs using the ping command prompt

16. Ping the default gateway from each PC and then PCs belonging to the other subnet

17. If successfully replies are recieved with no network is functioning

Result

Thus an internetwork using a wireless LAN in Packet Tracer and Internet cloud was successfully designed and configured in cisco packet tracer and connectivity was successfully verify

AIM

To design an

PROCEDURE

Step 1 B

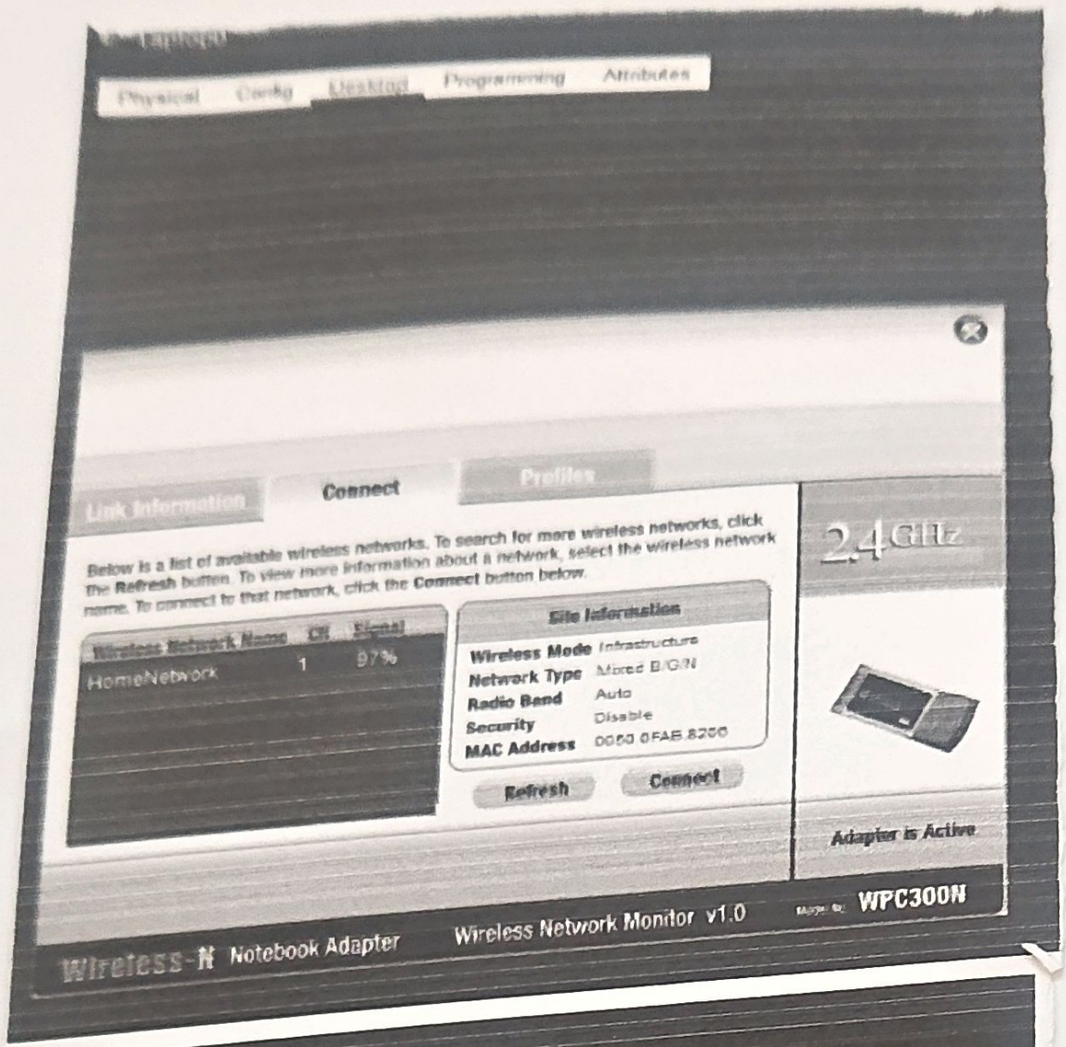
1.

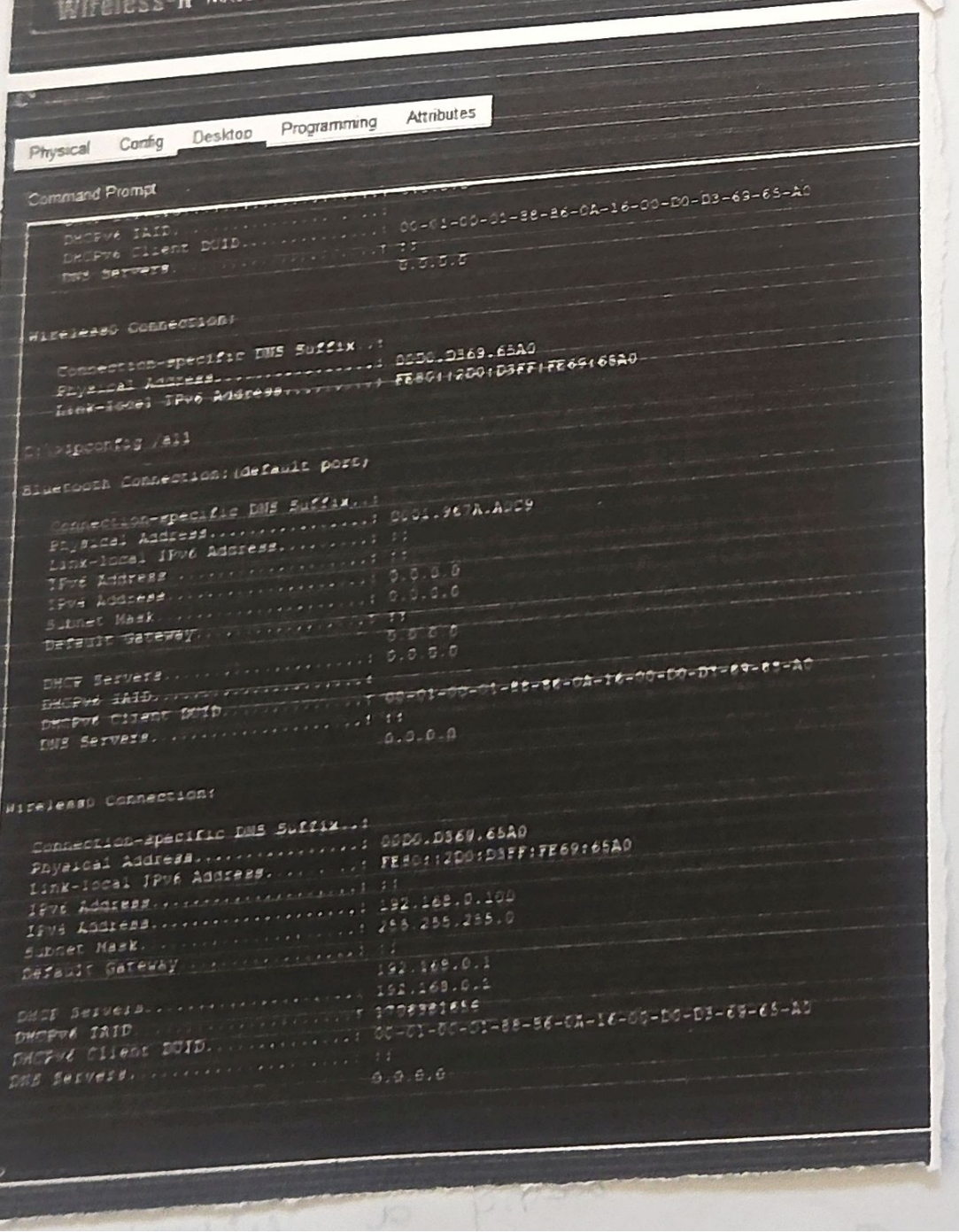
2.

3. Con

is t

4. Use


Physical
Config
Desktop
Programming
Attributes
Link Information
Connect
Profiles
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
CH
Signal
HomeNetwork
1
97%
Site Information
Wireless Mode Infrastructure
Network Type Mixed B/G/N
Radio Band Auto
Security Disable
MAC Address 0050 0FAB 8200
Refresh
Connect
2.4GHz
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
Model No. WPC300N


Physical
Config
Desktop
Programming
Attributes
Command Prompt
DHCPv6 IAID. . . . . : 00-01-00-01-88-86-0A-16-00-D0-D3-69-65-A0
DHCPv6 Client DUID. . . . . : 00-01-00-01-88-86-0A-16-00-D0-D3-69-65-A0
DNS Servers . . . . . : 0.0.0.0
Wireless0 Connection:
Connection-specific DNS Suffix . :
Physical Address. . . . . : 0060.D369.65A0
Link-local IPv6 Address. . . . . : FE80::2D0:D3FF:FE69:65A0
C:\>ipconfig /all
Bluetooth Connection: (default port)
Connection-specific DNS Suffix . :
Physical Address. . . . . : 0001.967A.AAC9
Link-local IPv6 Address. . . . . : ::
IPv6 Address . . . . . : 0.0.0.0
IPv4 Address. . . . . : 0.0.0.0
Subnet Mask . . . . . : ::
Default Gateway . . . . . : 0.0.0.0
0.0.0.0
DHCP Servers. . . . . :
DHCPv6 IAID. . . . . : 00-01-00-01-88-86-0A-16-00-D0-D3-69-65-A0
DHCPv6 Client DUID. . . . . : ::
DNS Servers. . . . . : 0.0.0.0
Wireless0 Connection:
Connection-specific DNS Suffix . :
Physical Address. . . . . : 0060.D369.65A0
Link-local IPv6 Address. . . . . : FE80::2D0:D3FF:FE69:65A0
IPv6 Address. . . . . : ::
IPv4 Address. . . . . : 192.168.0.100
Subnet Mask. . . . . : 255.255.255.0
Default Gateway . . . . . : ::
192.168.0.1
DHCP Servers. . . . . : 192.168.0.1
DHCPv6 IAID . . . . . : 1708381656
DHCPv6 Client DUID. . . . . : 00-01-00-01-88-86-0A-16-00-D0-D3-69-65-A0
DNS Servers. . . . . : ::
0.0.0.0

EX NO: 10b Design and configuration of an Internetwork using wireless router DHCP Server and Internet cloud.

AIM

To design and configure an internetwork using a wireless router. DHCP server and internet cloud in CISCO packet Tracer, and verify connectivity between the PC and the cisco.com server the simulate Packet transmission between subnetworks without a router and observe the communication between the subnets.

PROCEDURE:

Step.1 Build the network topology

1. Open CISCO packet tracer.
2. Add the required devices.
   - → Wireless router.
   - → PC
   - → Laptop
   - → Cable modem.
   - → Internet cloud.
   - → CISCO.com server.
3. Connect the devices according to the topology is the manual
4. Use: → Copper straight-through cable between PC and wireless rates.
   - → Copper straight-through cable b/w wireless routes and cable median.
   - → Coaxial cable b/w cable medium and Internet cable.
   - → Copper straight through cable b/w Internet cloud and cisco.com server.

Step 2: configure the wireless router

1. Open the wireless router, Select GUI ⇒ wireless.
2. Set the network name (SSID) to. Home Network
3. Go to the setup tab and verify the DHCP server is enabled.
4. Configure the DNS server address as 208.67.220.220
5. Save the settings.

Step 3 Configure the laptop

1. Open the laptop and go to the Physical tab
2. Power off the laptop and remove the ethernet copper module.
3. Install the wireless WPC300N module. and power on the laptop
4. Go to: Desktop ⇒ PC wireless.
   → Select the connect tab connect to the home network wireless network

Step 4: Configure the PC using DHCP

1. Open the PC
2. Select Desktop → IP configuration.
3. Select the DHCP option.
4. The PC recievers its IPV4 configuration automatically from the wireless router.
5. Open cmd, ipconfig/all.

6. Verify that the PC receivers an IP address in the 192.168.0.x range.

Steps: Configure the Internet cloud.

1. Open the Internet cloud and go to the physical tab.

2. Install the required cloud modular if they are not already present.

-> PT - Cloud - NM-1CX

-> PT - Cloud - NM-1CFE

3. Go to config - Cable and select from port casual to port Ethernet.

4. click add and under Ethernet Interface Select cable as the provides Network.

Step 6: Configure cisco.com Server as DHCP server

1. Open the cisco.com server.

2. Select the services tab.

3. Select DHCP

4. Turn the DHCP service on and configure the DHCP pool

Step 7: Configure DNS server.

-> Selad DNS under services and turn DNS on

-> Enter: Name : cisco.com

Type : A record.

address : 208.67.220.220.

-> click ada

Step 8: Configure Global settings.

-> Select config -> settings and select static

-> Set: Gateway: 208.67.220.1

DNS Server: 208.67.220.220

Step 9. configure fast Ethernet 0

-> Select Config -> Fast Ethernet 0.

-> Select static

-> Enter: Ip address: 203.67.220.220

Subnet mask: 255.255.255.0

Step 10 configure Fast Ethernet 0

->Select config-> Fast Ethernet 0 and extend static

->Enter: Ip address: 208.67.220.220

Subnet mask: 255.255.255.0

Step 11. Verify connectivity.

->Refresh Pc IPV4 settings.

-> Open the PC

-> Go to Desktop -> cmd prompt

-> Enter:

ipconfig /release

Ipconfig /renew

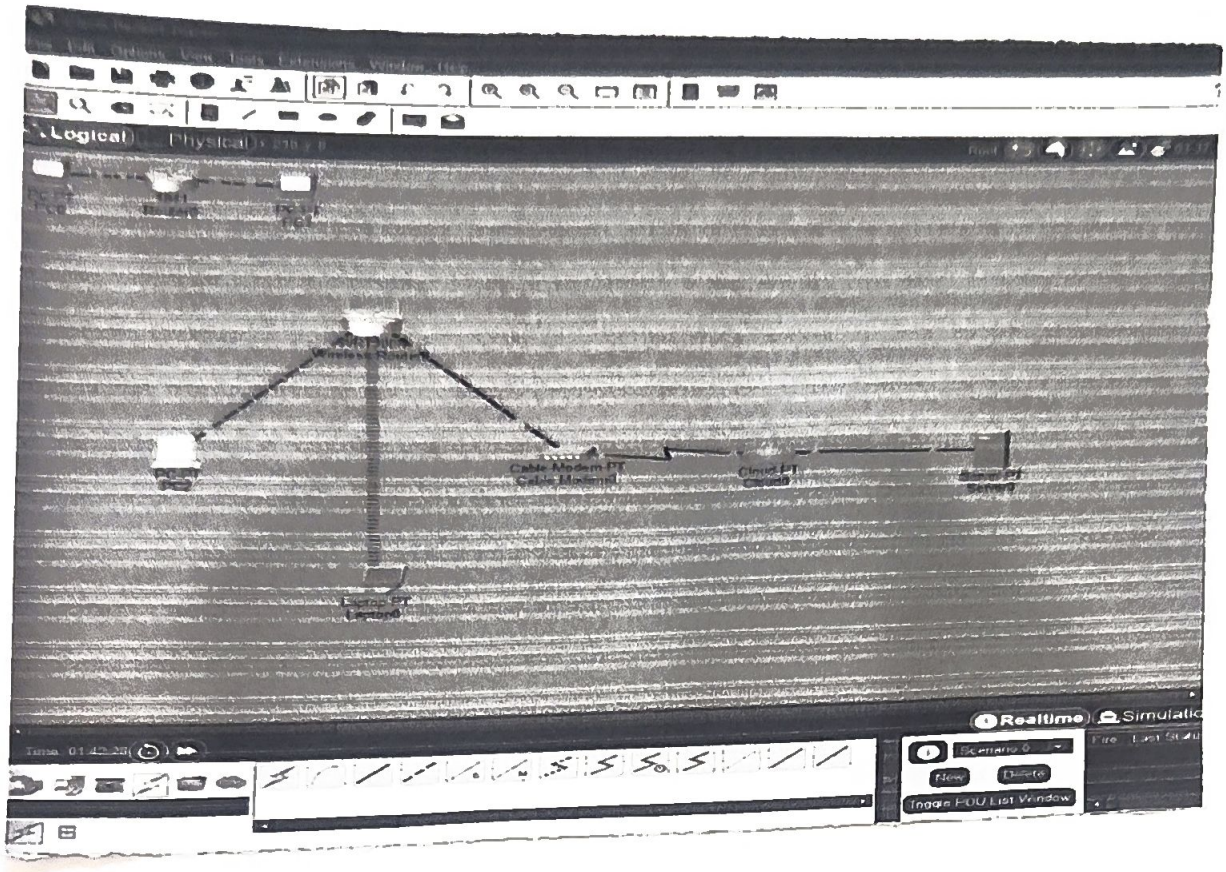
Step 12: Test cisco.com connectivity.

In cmd ether.

-> ping cisco.com.

-> wait for Ito response

-> Verify that 4 replies are recieved.



Result:

Thus the internetwork using a wireless router. DHCP server and Internet cloud was successfully designed and configured in cisco packet tracer and connectivity was successfully recieved.



10. (c) Design and configuration of a Internetwork in the lab using Switches, Routers and Ethernet cables.

AIM:

To implement classless IP subnetting cisco Packet Tracers and establish communication between different subnetworks using appropriate IP addresses and subnet masks.

PROCEDURE:

1. Open cisco packet tracer and establish communication between different subnetworks using appropriate IP addresses and subnet mables.

2. Create the topology in cisco packet Tracer.

2 - routers → R_1 & R_2, 2 switches → S_1 & S_2 and

1. PC's → PC_1 → PC_2, PC_1 - PC_5 → S_1, S_1 → R_1, PC_6 - PC_{10} → S_2, S_2 → R_2

4. Connect Routers and Router 2 using a ethernet cable.

5. Use the network address 192.168.1.0/24 and divide it into subnets using the /24 subnet mask (255.255.255.254)

6. Assign suitable IP addresses and Subnet masks to the Pc 3 in the 1st and the 2nd Subnet

7. Configure the IP addresses for the corresponding interfaces of R_1 and R_2

8. Enable all the routes interfaces using the no command shutdown command.

9. Configure the required static routes in R_1 and R_2 to enable communication between the 2 subnets

10. Verify the configured interfaces using the show ip interface brief command.

11. Test the connectivity between the PP_3 using the Ping command.

12. Verify the successfully ping replies are recieved confirming communication between the different subnetworks.

Result :

Thus a simple Internetwork using a router was successfully designed and configured in cisco packet tracer and communication between the connected network was successfully established and verified.